UK Macroeconomic Impact on FTSE 100 (2023–2025)

An Examination of Market Trends, Inflation, and Interest Rate Dynamics

Abstract

This report examines the relationship between FTSE 100 market performance and key macroeconomic indicators in the United Kingdom. The analysis integrates market price data, reference information from the London Stock Exchange, and macroeconomic indicators, including inflation and interest rates.

The datasets were consolidated into a unified analytical framework to enable time-series analysis. The report is organised into sections that examine market performance, inflation dynamics, and the monetary policy environment to provide a structured understanding of the economic factors influencing the FTSE 100.

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1. Introduction

Equity markets do not exist in isolation; they are constantly influenced by broader economic forces such as inflation, interest rates, and overall economic growth. Understanding these interactions is essential for investors, analysts, and policymakers seeking to anticipate market behaviour and manage risk effectively.

This report examines the UK macroeconomic impact on the FTSE 100 between 2023 and 2025, integrating three key data sources:

1. **FTSE 100 Market Data:** Historical price, open, high, low, trading volume, and percentage change — representing equity market performance.

2. **Inflation Indicators:** Consumer Price Index (CPI), CPIH, and Retail Price Index (RPI) — reflecting inflation pressure in the UK economy.

3. **Bank Rate:** Monthly average of the Bank of England's official interest rate — reflecting monetary policy stance.

1.1 Relevance

Equity markets are sensitive to macroeconomic conditions:

- Inflation affects corporate costs, consumer spending, and asset valuations.
- Interest rate adjustments influence liquidity, borrowing costs, and investor risk appetite.

By analysing these indicators together, this report identifies the patterns, relationships, and insights necessary to understand market dynamics and potential future trends.

1.2 Objective

The primary objective of this report is to evaluate the relationship between macroeconomic indicators (inflation and interest rates) and FTSE 100 performance over the specified period. Specifically, it aims to:

- Track trends in market performance and volatility
- Identify periods of inflation acceleration or divergence across measures
- Examine the timing and magnitude of monetary policy responses
- Understand how macroeconomic shifts correlate with equity market returns

2. Data Consolidation & Alignment

This section outlines the preparation and structuring of the integrated dataset, ensuring consistency, temporal alignment, and analytical readiness for subsequent market and macroeconomic assessment. The three core data sources were used to construct a unified monthly analytical dataset.

Daily FTSE 100 trading data was first aggregated into monthly performance metrics and subsequently integrated with company classification data and key UK macroeconomic indicators. This structured dataset enables simultaneous evaluation of market performance, sector dynamics, and macroeconomic conditions within a single analytical framework.

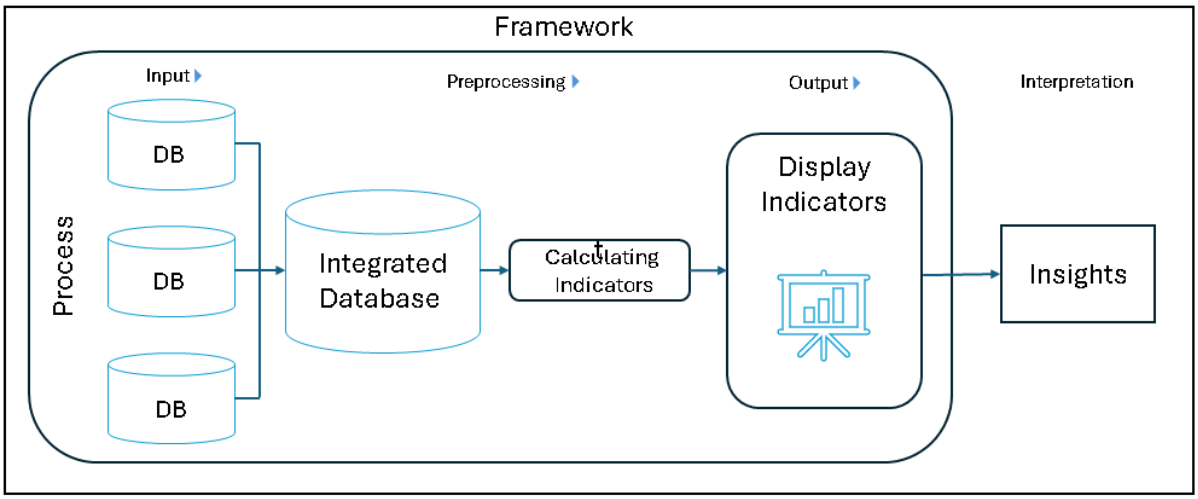

Figure 1. Integrated Monthly Analytical Framework

This integrated structure forms the foundation, while consolidation established structural consistency across datasets, analytical clarity required further temporal refinement to ensure alignment between market performance and macroeconomic idicators.

2.1 Temporal Aggregation Strategy

The original FTSE 100 dataset consisted of daily trading observations reflecting short-term volatility embedded within daily market observations, as illustrated below,

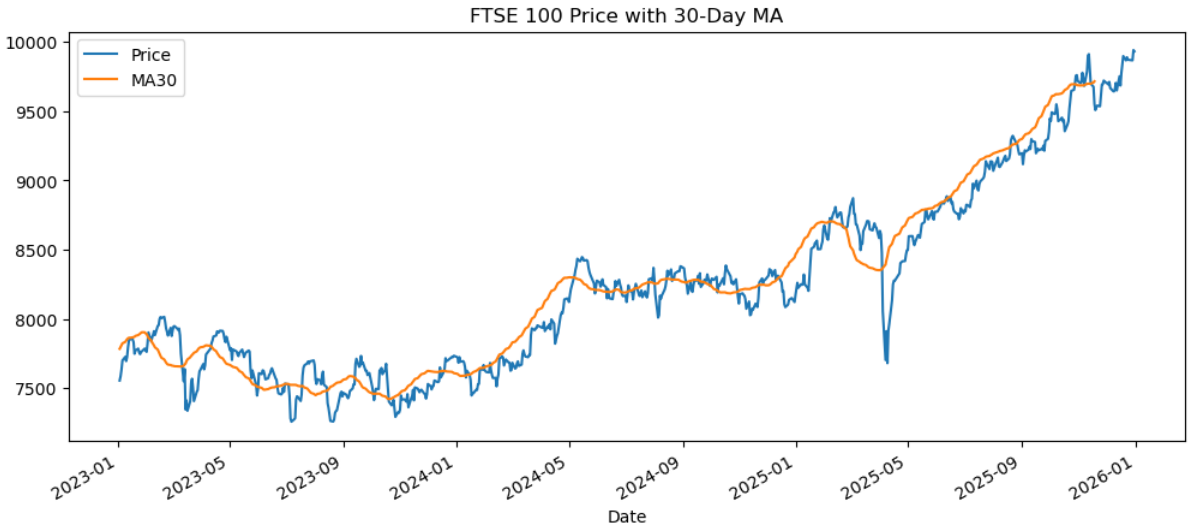

Figure 2.FTSE 100 Daily Closing Price with 30-Day Moving Average

The moving average smooths short-term price movements, revealing the underlying market direction and highlighting high-frequency volatility in daily observations. Consequently, daily data was aggregated into monthly performance metrics for subsequent analysis. To ensure analytical coherence and comparability across datasets, daily market data was aggregated into monthly performance metrics.

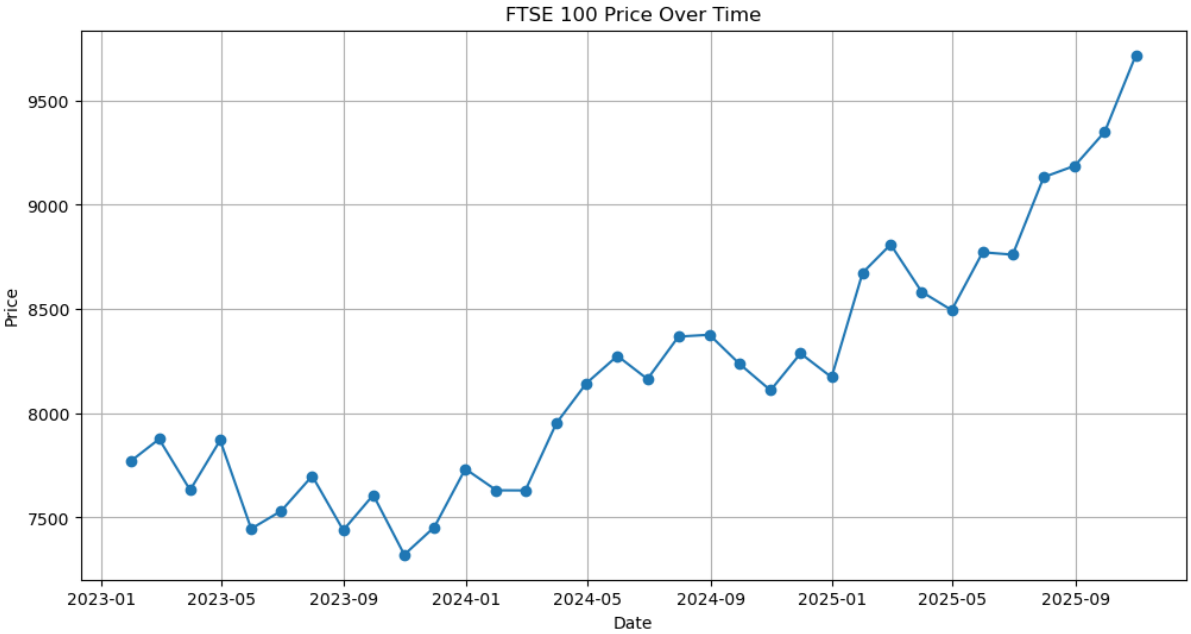

Figure 3. FTSE 100 quarterly Closing Price

Monthly aggregation reduces short-term volatility and highlights broader market cycles, providing a clearer basis for macroeconomic alignment and strategic trend analysis.

2.2 Analytical Variable Structure

Construction of variables for market and macroeconomic analysis	
Market Outcomes	Economic Drivers
• Monthly Closing Price • Monthly Return (%) • Volume	• CPIH Change • CPI Change • RPI Change • Bank Rate

Following data consolidation and temporal alignment, the integrated dataset was organised into two key analytical groups. Market performance variables capture the behaviour of the FTSE 100 through monthly price movements, returns, and trading activity. In parallel, macroeconomic variables represent the broader economic environment, including inflation indicators and the Bank Rate. Structuring the dataset in this way creates a clear link between market outcomes and external economic drivers, providing the foundation for the analysis presented in the following sections.

3. Analysis

3.1 Market Trend

With the structured monthly dataset established, the analysis begins by examining the overall behaviour of the FTSE 100 index over the study period.

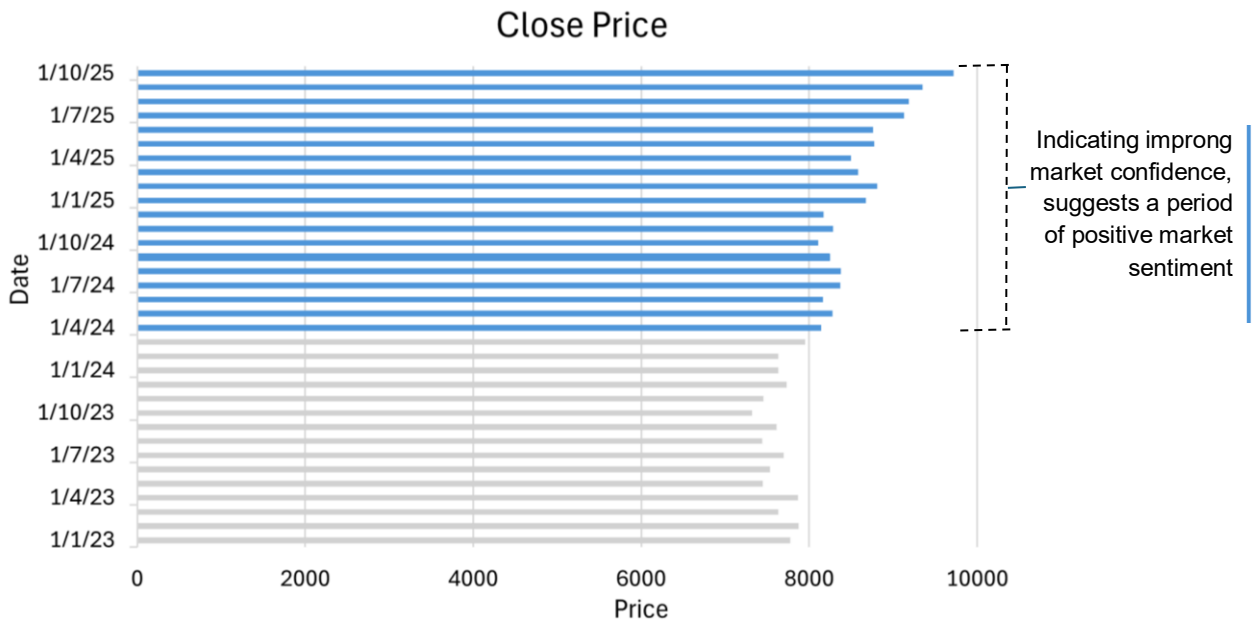

Figure 4. Close price trend

The index exhibits moderate variation, reflecting normal market adjustments and investor reactions to economic conditions.

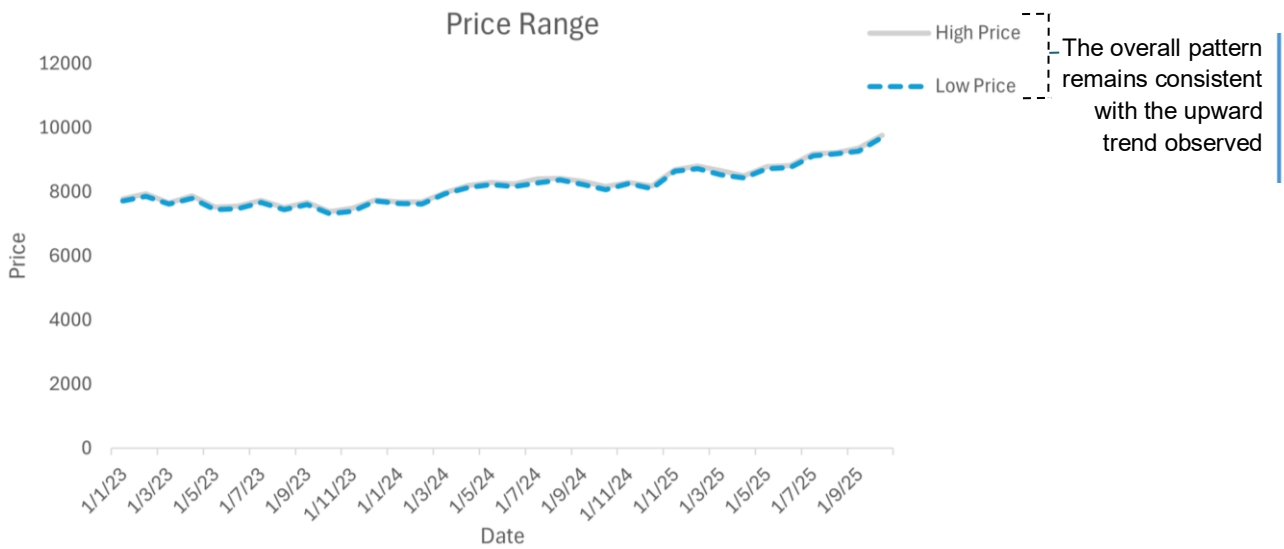

Figure 5. Quarterly price range

Overall, this indicates that market expansion occurred alongside manageable levels of volatility.

3.2 Inflation Environment

The index shows a gradual recovery from a slightly negative average return in 2023 to consistent positive growth in subsequent years. Despite the increase in index levels, market volatility remained relatively stable, indicating that the growth occurred without excessive instability.

Year	Year-End Index Level	Avg FTSE Change	FTSE Volatility	Latest CPI Change	Latest Bank Rate
2025	9,717.25	0.04	0.51	3.60	2.78
2024	8,173.02	0.03	0.56	2.50	2.79
2023	7,733.24	-0.03	0.49	4.00	2.63
Total	**9,717.25**	**0.01**	**0.52**	**3.60**	**2.78**

Table 1. Summary of Market Performance

The negative average change recorded in 2023 reflects a phase of market uncertainty, while the positive growth rates in 2024 and 2025 indicate a clear recovery and strengthening of investor confidence. The highest index level observed in 2025 further confirms the market's sustained upward movement. Overall, the indicators suggest that there is an initial instability toward a more stable and expanding market environment by the end of the period.

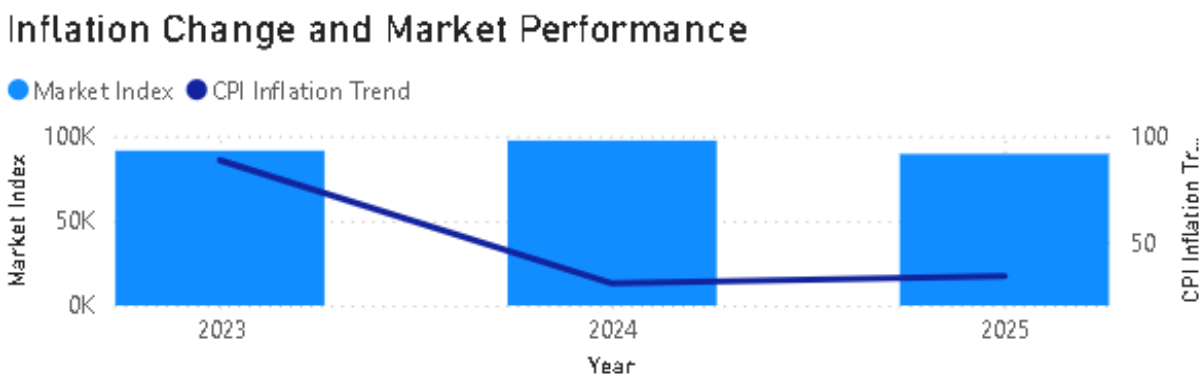

Figure 6. CPI Inflation Change (%) vs FTSE 100 Index

The chart illustrates the CPI inflation rate over the observation period. Inflation levels show noticeable fluctuations, with higher inflation pressures in the earlier years followed by a gradual decline. The reduction in inflation indicates improving price stability in the economy, which may contribute to a more favourable environment for equity market performance.

3.3 Inflation Trend

To examine the macroeconomic environment influencing financial markets. Inflation is a key indicator of economic stability, as it affects consumer purchasing power, corporate costs, and monetary policy decisions. Analysing the behaviour of inflation indicators such as CPI, CPIH, and RPI provides insight into the level of price pressures within the UK economy during the observation period.

Year	CPI_change	CPIH_change	RPI_change	Latest_CPI_RPI_Diff
2024	30.40	39.40	43.20	-1.00
2025	33.90	39.40	41.60	-0.70
2023	88.60	82.00	117.70	-1.20
Total	**152.90**	**160.80**	**202.50**	**-0.70**

Table 2.Summary of UK Inflation Indicators

The highlighted year 2023 represents the period with the highest inflation pressure across all indicators. Both CPI and RPI changes reached significantly higher levels compared with the following years, indicating a strong surge in price increases within the economy. Overall, the data indicate that inflation moved from a period of elevated pressure in 2023 toward a more stable inflation environment in the subsequent years.

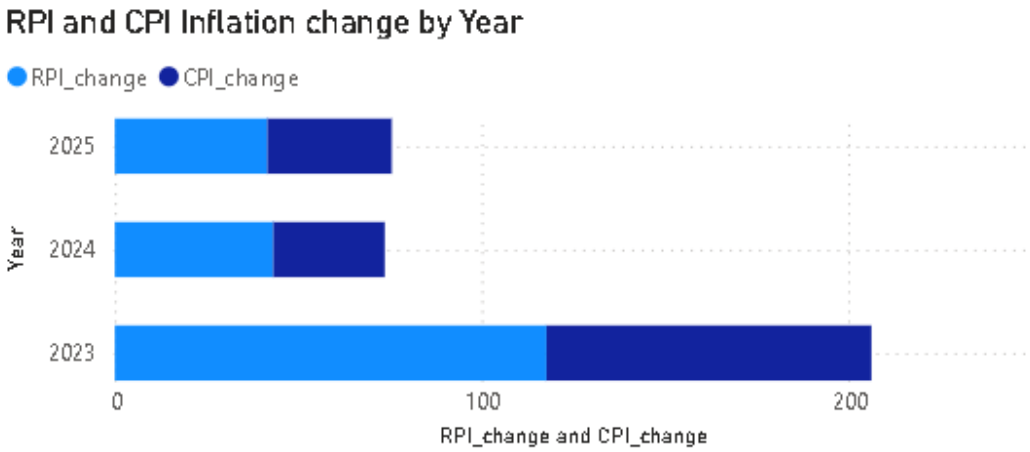

Figure 7. Change in Inflation

This indicates a period of strong price increases within the economy. In contrast, inflation declined sharply in 2024, reflecting a substantial easing of price pressures. Although a slight increase is observed in 2025, inflation levels remained considerably lower than the peak recorded in 2023. Overall, the visual confirms that inflation moved from a period of elevated pressure toward a more stabilised environment over time.

3.4 Interest Rate Environment

Following the analysis of inflation trends, it is important to examine the role of monetary policy in responding to changing economic conditions. Analysing the movement of interest rates alongside inflation indicators provides insight into how monetary policy may influence financial markets and broader economic activity.

Year	Sum of Bank_Rate	Latest Bank Rate	Avg CPI Change	Latest CPI Change
2023	3261.00%	2.63	7.38	4.00
2024	3275.00%	2.79	2.53	2.50
2025	2778.00%	2.78	3.39	3.60
Total	**9314.00%**	**2.78**	**4.50**	**3.60**

Table 3. Bank rate and Inflation indicator

The results indicate that the total Bank Rate value is highest in 2023 compared with the following years. This suggests that monetary policy conditions were relatively tighter during that period, likely reflecting the central bank's response to elevated inflation pressures in the economy. Higher interest rates are typically used to control inflation by reducing borrowing and moderating spending. In subsequent years, the Bank Rate shows only minor variations, indicating a relatively stable monetary policy environment as inflation pressures began to ease.

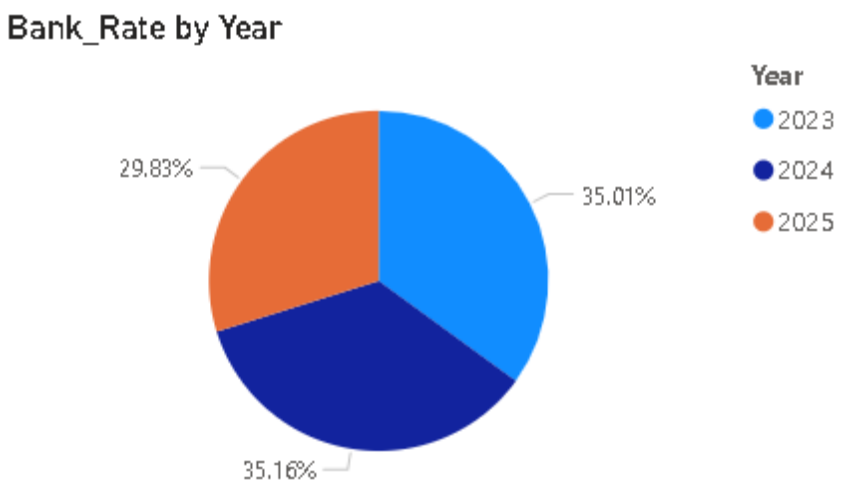

Figure 8. Annual distribution of Bank Rate

The chart indicates that the Bank Rate remained relatively consistent over the three years, reflecting a cautious and stable monetary policy approach aimed at managing inflation while maintaining economic stability.

Additional analysis indicates that the Bank Rate fluctuated within a relatively narrow range, with an average rate of approximately 2.73 and values ranging between 2.60 and 2.81, further confirming the overall stability of monetary policy during the period.

3.5 Market and Macroeconomic Interaction

The analysis of market performance, inflation indicators, and interest rates provides a broader understanding of the economic environment influencing the FTSE 100. The market trend shows a gradual recovery and growth from 2023 to 2025, reflecting improving investor confidence. At the same time, inflation indicators reveal a significant decline after the peak observed in 2023, suggesting easing price pressures within the economy. The Bank Rate remained relatively stable throughout the period, indicating a cautious monetary policy response aimed at maintaining economic stability while controlling inflation. Together, these indicators suggest that the moderation of inflation and the stable interest rate environment contributed to a more favourable climate for equity market performance.

4. Conclusion

This analysis examined the relationship between FTSE 100 market performance and key macroeconomic indicators, including inflation and interest rates. The results show that the FTSE 100 experienced steady growth during the observation period despite moderate fluctuations. Inflation pressures peaked in 2023 before gradually declining in the following years, indicating improving price stability. Meanwhile, the Bank Rate remained relatively stable, reflecting a cautious monetary policy stance. Overall, the findings suggest that a stabilising macroeconomic environment supported improved market performance during the later stages of the period.